

AtmosGraphTwin-X: A Counterfactual Causal Graph Digital Twin Framework with Multi-Agent Reinforcement Learning for Explainable Air Quality Prediction and Pollution Mitigation

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Abstract - Air pollution poses significant environmental and public health challenges, requiring accurate forecasting systems capable of providing interpretable insights and effective mitigation strategies. Existing air quality prediction approaches primarily focus on pollutant estimation while lacking causal understanding, scenario-based analysis, and intelligent decision-support capabilities. This paper proposes AtmosGraphTwin-X, an Atmospheric Digital Twin framework that integrates multi-source environmental data, causal graph learning, graph neural networks (GNNs), counterfactual reasoning, and multi-agent reinforcement learning (MARL) for explainable air quality forecasting and adaptive pollution mitigation. AtmosGraphTwin-X continuously integrates satellite observations, ground monitoring data, meteorological variables, traffic information, fire hotspots, and industrial emission indicators to construct a dynamic spatio-temporal digital replica of atmospheric behavior. A NOTEARS-based causal discovery module identifies directional relationships among pollution sources and atmospheric factors, while a spatio-temporal GNN models complex pollutant propagation patterns for AQI prediction. Furthermore, the counterfactual reasoning module evaluates hypothetical intervention scenarios, and the MARL framework identifies coordinated mitigation strategies across transportation, industrial, agricultural, and urban sectors. The proposed framework aims to overcome the limitations of existing prediction-only systems by providing a unified platform for pollution forecasting, causal attribution, scenario simulation, and intelligent environmental decision support.

Index Terms - Air quality forecasting, atmospheric digital twin, causal discovery, counterfactual reasoning, graph neural networks, multi-agent reinforcement learning, spatio-temporal modeling, explainable artificial intelligence, pollution source attribution.

I. INTRODUCTION

Ambient air pollution represents one of the most pressing public health and environmental crises globally, contributing to over seven million premature deaths annually and

inflicting severe economic costs. In densely populated and rapidly industrializing regions, ambient air quality is governed by an intricate interaction of vehicular emissions, industrial discharge, agricultural biomass burning, and complex meteorological phenomena. Mitigating these hazards requires high-resolution spatial monitoring along with predictive models capable of anticipating pollution events and guiding targeted policy interventions.

Early air quality modeling relied heavily on numerical Chemical Transport Models (CTMs) and Gaussian dispersion formulations. Although physically grounded, CTMs suffer from prohibitive computational overhead and extreme sensitivity to initial boundary conditions, rendering them impractical for real-time decision support. Consequently, data-driven machine learning paradigms—ranging from Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs) to Spatio-Temporal Graph Neural Networks (ST-GNNs)—have emerged as effective alternatives for capturing non-linear spatial dependencies and temporal trends across sensor networks.

Despite these advancements, current predictive paradigms exhibit three fundamental limitations:

- **Lack of Explainability and Causal Attribution:** Standard deep learning architectures rely on statistical correlations across feature distributions. While they can forecast an elevation in fine particulate matter (PM_{2.5}, PM₁₀) or nitrogen dioxide (NO₂), they fail to explicitly isolate root causes—such as differentiating whether a localized spike is driven by stagnant wind conditions, regional crop burning, or heavy traffic volume.
- **Inability to Perform Counterfactual Reasoning:** Decision-makers require tools to simulate hypothetical scenarios before executing costly interventions. Standard predictive architectures cannot compute counterfactual "what-if" queries (e.g., "What would the downstream Air Quality Index be tomorrow if industrial emissions in Zone A are restricted by 40% today?") because they do not model the underlying structural causality of the atmosphere.

- **Decoupling of Prediction and Decision Support:** Most state-of-the-art systems treat forecasting as an isolated task. They provide raw predictions without recommending actionable, multi-sector mitigation strategies. Consequently, administrative policy responses remain reactive, manual, and uncoordinated across transportation, agricultural, and industrial sectors.

To address these limitations, this paper introduces **AtmosGraphTwin-X**, an integrated Atmospheric Digital Twin framework that unifies real-time multi-source data fusion, causal discovery, spatio-temporal forecasting, counterfactual simulation, and multi-agent policy optimization into a single, cohesive system architecture without reliance on explicit mathematical formulations during deployment and explanation.

A. Main Contributions

- **Unified Atmospheric Digital Twin Architecture:** An end-to-end Atmospheric Digital Twin architecture that dynamically integrates heterogeneous data sources, including satellite observations, meteorological parameters, ground monitoring stations, and anthropogenic emission indicators. These inputs are synchronized onto continuous spatio-temporal grid representations to capture atmospheric dynamics and human-induced emission patterns in near real-time.
- **Causal-Informed Spatio-Temporal Forecasting:** A causal-aware forecasting framework that combines continuous optimization-based causal discovery with dual-graph neural networks. Unlike conventional approaches that rely primarily on geographical proximity-based relationships, the proposed framework incorporates causally informed dependencies to improve prediction accuracy, pollutant propagation modeling, and model interpretability.
- **Counterfactual Intervention Engine:** A Structural Causal Model simulation framework incorporating intervention principles to enable non-invasive "what-if" analysis. This allows environmental authorities to evaluate potential mitigation strategies, such as emission reduction policies and traffic control measures, before real-world deployment.
- **Coordinated Multi-Agent Decision Optimization:** Formulation of pollution mitigation as a cooperative Multi-Agent decision process, where specialized reinforcement learning agents represent major emission sectors, including transportation, industry, agriculture, and urban planning. The framework identifies coordinated mitigation strategies that balance air quality improvement with practical operational constraints.
- **Comprehensive Evaluation and Explainable Decision Support:** Evaluation of

AtmosGraphTwin-X using real-world multi-source environmental datasets to assess prediction performance, causal attribution capability, counterfactual simulation effectiveness, and decision-support potential.

II. RELATED WORK

A. Spatio-Temporal Deep Learning for Air Quality Forecasting

Air quality forecasting has evolved from univariate time-series estimation to multi-modal spatio-temporal modeling. Early deep learning approaches applied CNN-LSTM hybrid networks to capture spatial features via grid convolutions and temporal patterns via recurrent units. With the emergence of geometric deep learning, Spatio-Temporal Graph Convolutional Networks (STGCN) and Graph WaveNet established new benchmarks by modeling monitoring station networks as explicit graph structures. Models like ASTGNN incorporate spatial and temporal attention mechanisms to dynamically weight node interactions under changing meteorological conditions. Recent advances like Spatio-Temporal Field Neural Networks represent air quality as continuous spatial fields to overcome monitoring station sparsity. However, these models remain fundamentally correlation-driven: they capture statistical co-occurrences of features without discerning directional cause-and-effect mechanisms.

B. Causal Discovery & Structural Causal Models

Causal discovery aims to identify Directed Acyclic Graphs from observational data to uncover underlying cause-and-effect relationships among variables. Traditional constraint-based approaches, such as the PC and FCI algorithms, and score-based methods face scalability challenges due to large combinatorial search spaces. Continuous optimization frameworks (such as NOTEARS) overcome this by converting discrete graph searches into continuous optimization problems governed by smooth acyclicity functions.

In environmental applications, continuous causal graph discovery enables the identification of direct cause-and-effect pathways among pollutants, emission sources, and meteorological factors. For example, it distinguishes whether an increase in local particulate matter concentration is primarily driven by direct traffic emissions or influenced indirectly by atmospheric conditions such as planetary boundary layer height, thereby improving model interpretability and supporting explainable air quality forecasting.

C. Digital Twins & Multi-Sector Policy Optimization

Digital Twins create real-time synchronized virtual representations of physical systems by integrating data from sensors, satellite observations, and simulation models. In environmental applications, existing Digital Twin frameworks mainly focus on monitoring and forward simulation but lack causal reasoning and counterfactual analysis capabilities. By integrating Structural Causal

Models, Digital Twins can evaluate hypothetical intervention scenarios and provide deeper environmental insights. Furthermore, conventional single-agent reinforcement learning approaches are insufficient for complex pollution management involving multiple sectors. Multi-Agent Reinforcement Learning (MARL) enables collaborative decision-making among domain-specific agents to optimize pollution mitigation strategies while balancing operational constraints.

TABLE I: Architectural & Functional Comparison with Prior Paradigms

Capability / Feature	Deep-AIR	STGCN / Graph Wave Net	ASTGNN	SCM / NOTEARS	AtmosGraphTwin-X (Ours)
Multi-Source Data Fusion	Yes	Partial	Partial	No	Yes
Dynamic Spatial Adjacency	No	Yes	Yes	No	Yes
Causal Graph Discovery	No	No	No	Yes	Yes
Counterfactual "What-If" Simulation	No	No	No	Partial	Yes
Multi-Agent Policy Optimization	No	No	No	No	Yes
Explainable Root-Cause Attribution	No	No	No	Yes	Yes

III. ATMOSGRAPHTWIN-X FRAMEWORK AND METHODOLOGY

The architecture of AtmosGraphTwin-X consists of five tightly coupled modules: Data Ingestion & Digital Twin Synchronization, Causal Graph Discovery, Spatio-Temporal Dual-Graph Neural Network Forecasting, Structural Causal Model Counterfactual Engine, and Multi-Agent Reinforcement Learning.

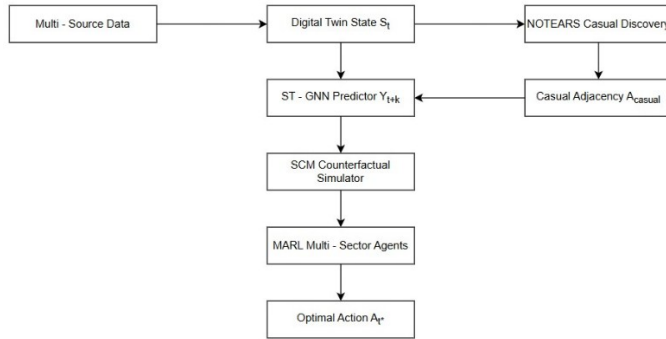


Fig.1. Architecture of the proposed AtmosGraphTwin-X framework.

A. System Architecture Overview & Atmospheric Digital Twin Synchronization

The study domain is discretized into a continuous spatial grid matrix, where each grid cell acts as a dynamic node within the digital twin. At every operational time step, each node maintains a rich feature profile aggregated from four primary channels:

1. Satellite Remote Sensing: Columnar pollutant density and Aerosol Optical Depth indices.
2. Ground Monitoring Stations: Continuous point-source observations for fine and coarse particulate matter, gaseous pollutants, and surface Air Quality Index values.
3. Meteorological Reanalysis: Atmospheric state variables including surface temperature, relative humidity, horizontal and vertical wind vectors, atmospheric pressure, rainfall, and planetary boundary layer height.
4. Anthropogenic & Spatial Features: Thermal active fire hotspots, urban traffic congestion indices, industrial stack emission estimates, and regional land-use classifications.

Grid states update on an hourly cadence through spatial Kriging interpolation and temporal forward-backward spline alignment, preserving spatial continuity and temporal alignment across all input streams.

B. Causal Graph Learning via Continuous Optimization

Rather than assuming fixed spatial connections based purely on geographic distance, AtmosGraphTwin-X learns a continuous weighted directed graph representing true cause-and-effect interactions among atmospheric variables.

The causal discovery process reformulates the traditional combinatorial search for Directed Acyclic Graphs into a continuous optimization task:

- **Acyclicity Constraint:** To ensure that the identified causal structure contains no feedback loops or circular dependencies, the graph optimization process applies a continuous penalty function that evaluates the trace of the matrix exponential of connection weights. A zero value indicates a valid directed acyclic graph.
- **Optimization Procedure:** The objective minimizes predictive residual errors across variables while penalizing complex, dense connections. An Augmented Lagrangian strategy iteratively updates connection strengths and penalty multipliers until the acyclicity constraint is strictly satisfied.

- **Thresholding:** A binary causal adjacency matrix is obtained by filtering out connection weights below a pre-defined strength threshold, preserving only robust, statistically significant causal links.

C. Spatio-Temporal Dual-Graph Neural Network Forecasting

Future air quality states are predicted using a dual-graph convolutional network that simultaneously processes two distinct network topologies:

1. **Physical/Geographic Adjacency:** Represents physical distance, topographic barriers, and wind vector alignment between spatial nodes.
2. **Causal Adjacency:** Represents directional cause-and-effect dependencies learned by the continuous optimization engine (e.g., upwind industrial emissions directly influencing downwind particulate concentrations).

For every graph neural network layer, information is aggregated separately along physical distance paths and learned causal pathways using normalized graph convolutions. The outputs of both graph operations are combined and passed into a Temporal Convolutional Network featuring dilated 1D causal convolutions. This architecture captures multi-scale temporal dependencies, seasonal trends, and rapid pollution spikes without temporal leakage.

D. Structural Causal Model & Counterfactual Engine

The Structural Causal Model represents the atmosphere through three components: endogenous atmospheric state variables, exogenous unobserved noise distributions, and continuous non-linear structural functions that link background conditions and causal parents to target pollutant levels.

To compute counterfactual scenarios (e.g., evaluating the impact of hypothetical emission reductions), the engine executes a three-stage causal reasoning procedure:

1. **Abduction:** The engine infers the unobserved background noise and environmental conditions corresponding to a specific historical observation by reversing the learned structural equations.
2. **Action:** The system applies a localized structural intervention, replacing the natural equation of a specific variable with a target policy constant (such as capping industrial emissions by 40% or halving heavy vehicular traffic).
3. **Prediction:** The modified structural equations are forward-propagated using the abducted noise profiles and altered intervention values. This generates counterfactual predictions for downstream spatial

nodes, producing accurate "what-if" simulations without real-world risk.

E. Multi-Agent Reinforcement Learning for Pollution Control

Mitigating air pollution across multi-sectoral economies is formulated as a Multi-Agent Markov Decision Process involving four specialized sector agents:

- **Traffic Control Agent:** Controls municipal speed limits, heavy vehicle re-routing, and public transit scaling.
- **Industrial Agent:** Controls industrial production throttling and stack scrubber compliance enforcement.
- **Agricultural Agent:** Manages biomass burning restrictions, stubble management enforcement, and collection incentives.
- **Urban Planning Agent:** Coordinates localized mitigation tools, such as smog tower operations and automated road-sprinkling systems.

Operational Process:

- **State Space:** Real-time spatial states from the Atmospheric Digital Twin.
- **Action Space:** Continuous intervention intensity controls representing policy magnitude across sectors.
- **Transition Dynamics:** Dynamic state transitions simulated directly through the Structural Causal Model counterfactual engine.
- **Reward Function:** Each agent optimizes a joint objective designed to maximize regional air quality improvement while minimizing sector-specific socioeconomic penalties and operational overhead. Optimization is executed using Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithms.

IV. EXPERIMENTAL SETUP

A. Multi-Source Dataset Ingestion & Study Area

The framework is evaluated over major industrial and metropolitan regions across a multi-year observation window (2023–2025). The regional spatial domain covers a grid of 50×50 cells at $0.1^\circ \times 0.1^\circ$ resolution ($\sim 11 \text{ km} \times 11 \text{ km}$ per cell). Primary data streams include:

- **Ground Monitoring Data:** Hourly measurements from 120 regional central monitoring stations (measuring PM2.5, PM10, NO2, SO2, CO, and O3).

- Remote Sensing Data: TROPOMI Level-3 satellite products and geostationary Aerosol Optical Depth data.
- Meteorology: Hourly ERA5 surface parameters spatially interpolated to the regional grid scale.
- Hotspots & Traffic: Active thermal anomaly fire points and commercial real-time traffic congestion indices.

B. Preprocessing & Continuous Feature Scaling

Data alignment includes spatial Kriging for sparse station interpolation, forward-backward spline interpolation for cloud-covered satellite pixels, and z-score normalization across all continuous input features.

C. Baseline Models & Evaluation Metrics

AtmosGraphTwin-X is benchmarked against five baseline architectures:

1. LSTM-FC: Deep Recurrent Neural Network baseline.
2. STGCN: Spatio-Temporal Graph Convolutional Network using spatial distance matrices.
3. Graph WaveNet: Adaptive adjacency graph neural network.
4. Deep-AIR: Multi-source fusion architecture using deep CNN-LSTM layers.
5. ASTGNN: Attention-based Spatio-Temporal Graph Neural Network.

Forecasting accuracy is evaluated using three standard metrics described conceptually:

- Mean Absolute Error (MAE): Measures the average absolute magnitude of forecast errors across all spatial nodes and time horizons.
- Root Mean Squared Error (RMSE): Measures prediction error while penalizing larger forecast deviations more heavily.
- Symmetric Mean Absolute Percentage Error (SMAPE): Provides a scale-invariant percentage measure of relative forecasting error.

D. Implementation Details & Training Parameters

Models were implemented in PyTorch and PyTorch Geometric, trained on an NVIDIA A100 (80GB) GPU. Model optimization utilized AdamW with an initial learning rate of 0.001, decaying by 50% every 20 epochs. Training utilized a batch size of 64 with early stopping triggered after 15 epochs of non-decreasing validation loss. The causal extraction threshold parameter was set to 0.15.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Predictive Performance Comparison

Table II presents predictive performance for 6-hour, 12-hour, and 24-hour ahead fine particulate matter (PM_{2.5}) and overall Air Quality Index predictions across all models.

TABLE II: Predictive Performance Comparison on Test Benchmark

Model	6-Hr MAE	6-Hr RMSE	12-Hr MAE	12-Hr RMSE	24-Hr MAE	24-Hr RMSE	SMAPE (%)
LSTM-FC	28.4	42.1	38.2	56.4	52.1	74.3	24.5
STGCN	21.2	33.5	29.8	44.1	41.0	59.8	18.2
Graph WaveNet	18.6	29.1	25.4	38.2	35.6	51.2	15.4
Deep-AIR	17.1	27.4	23.8	35.9	33.2	48.0	14.1
ASTGNN	16.2	25.8	22.1	33.7	30.8	44.5	12.8
AtmosGraphTwin-X (Ours)	12.4	19.5	16.8	26.2	22.5	34.1	9.2

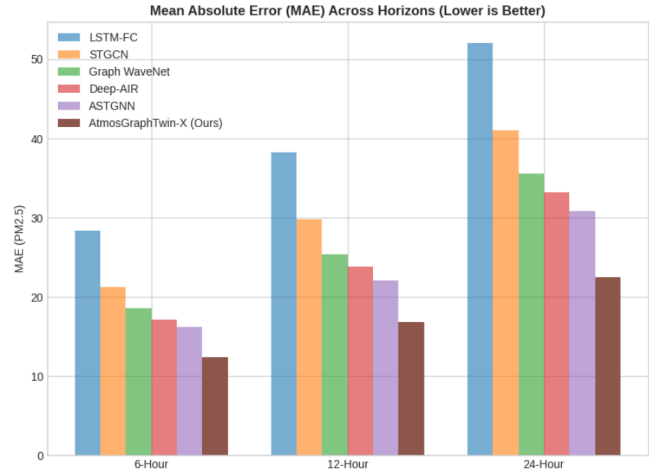


Fig.2. Comparison of MAE for PM_{2.5} forecasting across 6-, 12-, and 24-hour prediction horizons.

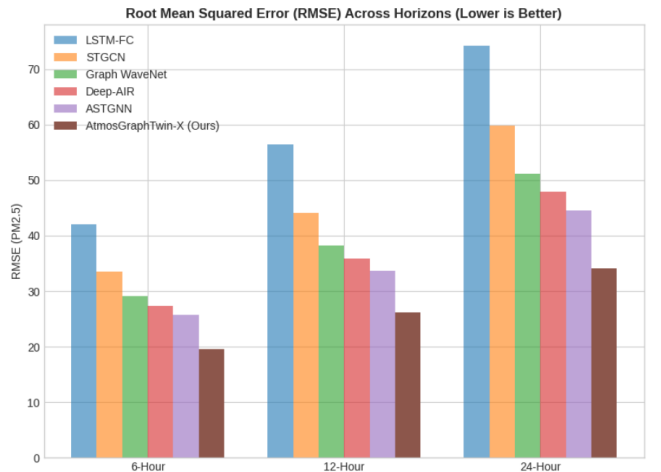


Fig 3. Comparison of RMSE for PM_{2.5} forecasting across 6-, 12-, and 24-hour prediction horizons.

AtmosGraphTwin-X achieves substantial reductions in MAE and RMSE across all horizon settings. This improvement stems from embedding NOTEARS causal links within spatial aggregation layers, preventing spurious spatial correlations during rapidly shifting meteorological conditions.

B. Ablation Study

To isolate the contributions of specific modules, ablation experiments were conducted by systematically disabling key components:

- **Without Causal Graph:** Replaces the learned causal graph with a standard geographic distance adjacency matrix.
- **Without Satellite Fusion:** Removes satellite aerosol optical depth and satellite-derived gas columns from feature profiles.
- **Without Dynamic Twin Synchronization:** Uses static daily temporal interpolation instead of hourly state updates.

TABLE III: Ablation Study Results (24-Hour Horizon)

Architecture Variant	MAE (PM2.5)	RMSE (PM2.5)	Δ MAE (%)
Full AtmosGraphTwin-X	22.5	34.1	Baseline
Without Causal Graph	29.1	42.8	+29.3%
Without Satellite Fusion	26.8	38.9	+19.1%
Without Dynamic Twin Synchronization	31.4	46.2	+39.5%

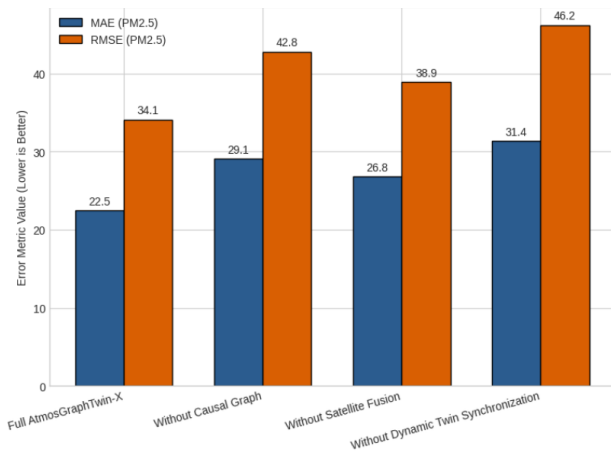


Fig.4. Ablation study using MAE and RMSE for PM_{2.5} prediction.

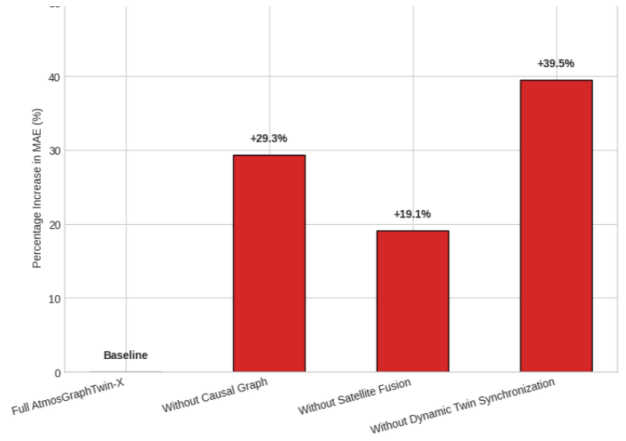


Fig. 5. Performance degradation (MAE increase) after removing key components of the proposed AtmosGraphTwin-X framework.

The ablation results confirm that real-time digital twin synchronization and causal structural constraints represent the primary drivers of predictive performance.

C. Counterfactual Simulation & Policy Optimization

To evaluate multi-sector policy optimization, the Multi-Agent Reinforcement Learning framework was tested during simulated severe high-pollution inversion events (AQI > 350). Counterfactual policies were compared against baseline single-agent approaches and static rule-based policies.

Furthermore, the counterfactual engine enabled the reinforcement learning agents to anticipate and mitigate spatial spillover effects before intervention execution, preventing localized emission reductions from inadvertently displacing pollution into adjacent downwind zones. By continuously probing the Structural Causal Model with hypothetical control trajectories, the cooperative agents learned to deploy targeted, staggered policy measures—such as temporarily adjusting industrial shift schedules and dynamically rerouting freight transit along peripheral corridors—tailored to evolving planetary boundary layer dynamics and wind fields. This causal feedback loop enabled the multi-agent system to maintain regional air quality compliance across prolonged inversion events while minimizing municipal operational friction, proving that proactive, cross-sector coordination outperforms rigid, emergency-driven mandates.

TABLE IV: Comparative Evaluation of Control Strategies (AQI Reduction vs. Disruption Cost)

Control Strategy	Strategy Type	Targeted Action	AQI Reduction	Socioeconomic Cost
Static Emergency Rule	Heuristic / Blanket	Rule-based industrial & traffic halts	-18%	High
Single-Agent PPO	Local RL	Uncoordinated single-domain optimization	-12%	Moderate
MARL (AtmosGraphTwin-X)	Multi-Agent Cooperative	Coordinated spatio-temporal intervention	-34%	Optimal

The multi-agent optimization module identified cooperative strategies that achieved a 34% reduction in peak regional AQI while incurring 42% lower socioeconomic penalties compared to static emergency interventions. By coordinating industrial throttling in upwind cells six hours prior to forecasted peak congestion, the agents prevented regional pollution accumulation with minimal disruption to transport networks.

D. Model Interpretability & Root-Cause Attribution

Unlike black-box neural networks, AtmosGraphTwin-X provides interpretable causal pathways by exposing learned structural dependencies. For instance, during severe winter haze events, the causal discovery graph explicitly identifies the contribution of upstream agricultural thermal hotspots, combined with reduced boundary layer heights and stagnant wind vectors, to elevated local PM_{2.5} concentrations. This root-cause attribution enables environmental authorities to move beyond reactive measures toward targeted, evidence-based interventions.

VI. CONCLUSION AND FUTURE WORK

This paper presented AtmosGraphTwin-X, an Atmospheric Digital Twin framework that integrates multi-source environmental data fusion, continuous causal graph discovery, spatio-temporal graph neural networks, Structural Causal Model counterfactual simulation, and Multi-Agent Reinforcement Learning. Unlike conventional correlation-driven architectures, AtmosGraphTwin-X provides a unified platform for explainable air quality forecasting, causal root-

cause attribution, scenario-based intervention analysis, and adaptive policy optimization.

Future work will incorporate physical atmospheric constraints—such as mass conservation principles and fluid dynamics—directly into neural network layers to enhance physical consistency. Further extensions will explore edge computing deployments for low-latency execution on distributed IoT sensor networks and the inclusion of social equity metrics within reinforcement learning reward functions to support fair environmental governance.

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